

Guided Generation of Computational Cognitive Models (GeCCo): A Comprehensive Analysis

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1 The Status Quo in Cognitive Science

The ultimate goal of cognitive science is to understand human intelligence by translating theories about the mind into formal, mathematical algorithms known as computational cognitive models. By fitting these models to human behavioral data, scientists can predict behavior and test competing psychological theories. However, the current pipeline for model discovery is fundamentally bottlenecked by human limitations. Translating a psychological theory into executable code is a slow, artisanal process requiring a rare blend of domain expertise, theoretical creativity, and software engineering skills. Furthermore, researchers inevitably carry biases based on their specific academic training. This theoretical blindness artificially shrinks the pool of possible models they consider, often causing them to miss simpler or more accurate explanations for human behavior.

2 The Paper’s Contribution

This paper introduces GeCCo, which stands for Guided generation of Computational Cognitive Models, as a novel, open-source pipeline that automates the scientific discovery process. The primary contribution of this work is proving that Large Language Models can act as

autonomous theoretical scientists. By feeding an artificial intelligence natural language task descriptions and raw behavioral data, GeCCo leverages the system to brainstorm hypotheses, write executable Python code to simulate them, and iteratively refine them based on mathematical feedback. Ultimately, GeCCo achieves state-of-the-art predictive accuracy while remaining strictly interpretable. This means it outputs readable code that human scientists can learn from to successfully democratize and accelerate cognitive modeling.

3 Methodology and Results

GeCCo operates on a rigorous ten-iteration loop to prevent the artificial intelligence from simply guessing or hallucinating. First, the model receives the rules of a psychological experiment, a sample of participant data, a blank Python template, and strict formatting guardrails. In each iteration, the model generates three distinct, competing cognitive models as Python functions. These models are then tested against a held-out dataset and scored using the Bayesian Information Criterion, which balances accuracy against complexity by rewarding simpler models. Finally, the winning model’s code and parameters are fed back into the system as a benchmark to beat in the next round.

When benchmarked against the best human-crafted models across four progressively complex cognitive domains, GeCCo consistently demonstrated superior performance. In decision making, it combined three separate human heuristics into a single, highly elegant equation. In the learning domain, it proposed a novel way to handle counterfactual learning while mirroring the natural decay of human memory. For planning tasks, the artificial intelligence built a unified theory of multi-step planning that actively bypassed a heavily debated two-system dichotomy in the existing literature. Finally, in working memory tasks, GeCCo correctly identified that cognitive load actively alters our deep reinforcement learning systems, which challenged a decade of psychological assumptions. Rigorous control tests further proved the system was genuinely reasoning without data contamination and established that the iterative feedback loop was the primary driver of its success.

4 Potential Criticisms

While groundbreaking, the GeCCo pipeline has several limitations that warrant scrutiny. Providing the language model with a structural Python template in the initial prompt might artificially restrict its creativity, subtly coercing it to only build models that resemble the starting template. Additionally, the system currently only receives a raw mathematical score as feedback. It knows if the numbers improved, but it does not receive qualitative feedback on whether its underlying psychological theory actually makes logical sense. The pipeline’s success is also entirely dependent on the reasoning capacities of external artificial intelligence models. If a base model is flawed, the output generated by GeCCo will inherently be flawed as well. Finally, the framework was tested on highly controlled, low-dimensional laboratory tasks, raising valid questions about how the pipeline will handle the chaotic and noisy nature of human behavior in real-world datasets.

5 Newer Research

Because GeCCo was introduced very recently, newer research primarily consists of concurrent projects that are integrating it into a broader vision of fully automated scientific discovery. Researchers are actively utilizing massive foundation models of human cognition to generate synthetic human data across thousands of potential experiments. Tools like GeCCo are acting as the necessary counterpart to these black-box models by taking generated data and reverse-engineering interpretable, human-readable rules from it. Concurrently, work is exploring the use of language models not just to build models, but to design psychological experiments from scratch. GeCCo represents the critical analysis half of a future pipeline where an artificial intelligence designs a test, simulates the data, and writes the theory to explain it without any human intervention.

6 Potential Extensions

The authors and the broader community have highlighted several exciting trajectories for extending this framework. One major extension involves integrating a second language model

into the feedback loop to act as a peer reviewer that critiques the psychological plausibility and parsimony of the code before it is tested mathematically. Researchers also aim to scale GeCCo out of simple tasks and apply it to complex, high-dimensional human functions like natural language processing, visual perception, and spatial navigation. Another potential direction is removing the starting Python templates entirely to see if the artificial intelligence can invent completely novel programming structures to explain human thought. Finally, deploying GeCCo on unstructured, naturalistic datasets such as smartphone usage patterns or driving telemetry could help discover models of human behavior outside the confines of the traditional laboratory.